

BANARAS HINDU UNIVERSITY
U. G. Semester II (NEP) Examination, 2024-25

COMPUTER SCIENCE

Paper No. CSCMD21 : Introduction to Information Technology

Time: 2:30 Hours

Full Marks: 70

REF NO. CECNF/N/2024-25/22

*Instructions: Attempt **five** questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate marks. Write your Roll No. at the top immediately on receipt of this question paper.*

Question 1. Choose the correct options:

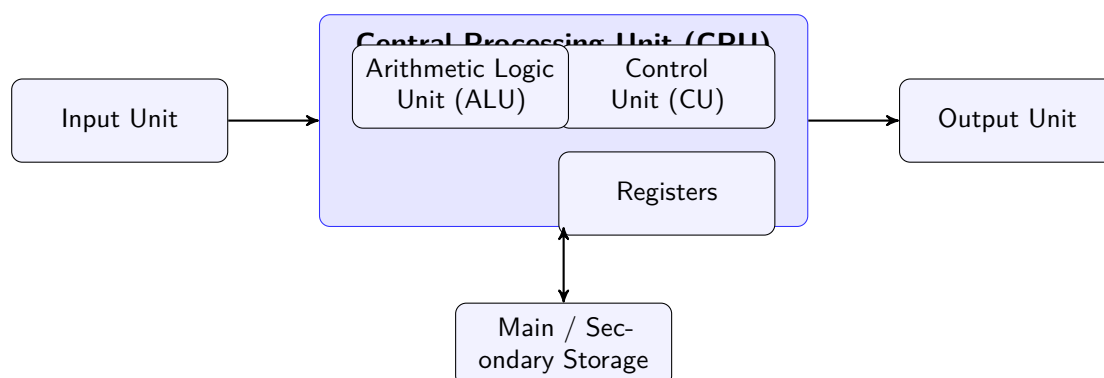
[7 × 2 = 14]

(a) Primary storage is:

- (i) cheap and volatile
- (ii) expensive and volatile
- (iii) cheap and non-volatile
- (iv) expensive and non-volatile

(b) Brain of computer system is:

- (i) ALU
- (ii) CU
- (iii) CPU
- (iv) Storage unit



(c) What is not correct regarding positional number systems?

- (i) The maximum value of a single digit is always equal to one less than the value of the base
- (ii) The value of each digit is determined by the digit itself

- (iii) Each symbol represents the same value regardless of its position in the number
- (iv) The value of each digit is determined by the base of the number system
- (d) $(952)_{10} = ?_8$
 - (i) 167
 - (ii) 1670
 - (iii) 761
 - (iv) 7610
- (e) What does 'bit' stands for in computer terms?
 - (i) Binary information term
 - (ii) Binary digit
 - (iii) Binary tree
 - (iv) All of the above
- (f) Registers are part of:
 - (i) main memory
 - (ii) secondary memory
 - (iii) CPU
 - (iv) ALU
- (g) Megabyte (MB) is equal to:
 - (i) $32 (2^5)$ bytes
 - (ii) $1024 (2^{10})$ bytes
 - (iii) $1,048,576 (2^{20})$ bytes
 - (iv) $1,073,741,824 (2^{30})$ bytes

Question 2.

- (a) (i) Find the complement of 6_8 .
 - (ii) Subtract 5610 from 9210 using complementary method. [7]
- (b) Discuss different types of ROMs in detail. [7]

Question 3.

- (a) Explain classification of commonly used secondary storage devices. [7]
- (b) How do we define magnetic tape storage capacity? What is the storage capacity of a magnetic tape if the tape density is 800 bpi and tape length is 600 feet? [7]

Question 4.

- (a) What do you understand about the storage capacity of a magnetic disk system? [7]
- (b) Write in detail on which parameters magnetic disk access time depends upon. [7]

Question 5.

- (a) Explain (i) relationship between hardware and software (ii) difference between software and Program. [7]
- (b) What do you understand about Database Management Systems (DBMS)? How is it different from file-oriented approach of organizing data? [7]

Question 6.

- (a) Explain software development steps in detail. [7]
- (b) Write the difference between (i) system software and application software (ii) compiler and interpreter. [7]

Question 7.

- (a) What do you understand about low-level and high-level languages? Write their advantages and limitations. [7]
- (b) What is the purpose of an operating system? Explain multiprogramming and multiprocessing techniques in detail. [7]